A Deep Learning Model for Green Algae Detection on SAR Images

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***Abstract*— This study developed a textural-enhanced deep learning (DL) model based on the classic U-net framework for green algae detection in Sentinel-1 synthetic aperture radar (SAR) imagery. Four special modifications are made in the framework: texture-fused input dataset, texture concatenation to effectively use the texture information, weighted loss function to settle the imbalance of algae–seawater samples, and an attention module to facilitate model focus on the discriminative features efficiently. To build the proposed model, we collected 119 Sentinel-1 SAR images acquired in the Yellow Sea and manually labeled 8441 samples, among which 4421/1896/2124 were used as the training/validation/testing dataset, respectively. Experiments show that the classification achieves the mean intersection over union (mIOU) of 86.31%, outperforming previous DL methods. Furthermore, each modification is effective, and the weighted loss function plays the most critical role. Moreover, we monitored green tide in the Yellow Sea from 2019 to 2021 using the proposed model and analyzed the relationship between green tide interannual variation and two primary environmental factors: nitrate concentration and sea surface temperature (SST). The interannual variation is characterized via three crucial indexes: bloom duration, coverage area, and nearshore damage. The detection results reveal that the bloom duration is the longest (shortest) in 2019 (2020), corresponding to the biggest (smallest) coverage area in 2019 (2020). In addition, the nearshore damage is the heaviest (lightest) in 2021 (2020). We also found that the interannual variation of green tide scales is partly related to the available nitrate concentration and SST variation in algae-distributed regions.**

***Index Terms*— Deep learning (DL), green algae, Sentinel-1 synthetic aperture radar (SAR) image, Yellow Sea.**

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I. Introduction

EXCESSIVE reproduction of green algae can cause the typical marine disaster: green tide in the coastal regions. Green tide comprises several green algae species, e.g., *Ulva prolifera*, *Ulva compressa* Linnaeus, *Ulva pertusa* Kjellman, and *Ulva linza* Linnaeus [1]. Ulva prolifera is the dominant species [2]. Green tide has appeared in many coastal areas, such as Florida Bay [3], the Gulf of Mexico [4], the Japan Sea [5], the Yellow Sea, and the East China Sea [6], threatening biodiversity, transportation, and tourism [7], [8]. Since 2007, the Yellow Sea (see Fig. 1) has become a typical disaster area that suffers from green tide every summer [9], drawing wide attention.

Remote sensing images, mainly optical and microwave synthetic aperture radar (SAR) images, have become the primary data sources for green algae detection. Satellite optical sensors usually have multiple spectral image channels, a wide swath, and a short revisit period. For example, Moderate Resolution Imaging Spectroradiometer (MODIS) and Geostationary Ocean Color Imager (GOCI) are widely used for green tide detection [10], [11]. The obstacle of optical images is that they cannot capture the sea surface under cloud cover conditions. On the contrary, SAR is an active radar, and its signals can penetrate clouds, rain, and fog. SAR sensors can also acquire the sea surface image day and night. The disadvantages of SAR images are narrow swath and long revisit periods. Therefore, optical and SAR images are supplementary instruments for green algae detection. Due to different observation objectives, the ability of different images to observe green algae varied. Images with low spatial resolution underestimate the algae coverage area [12]. Moreover, under the same spatial resolution and clear atmospheric conditions, optical images would detect more algae than SAR images due to the stronger penetrability to seawater [13].

Accompanied by the emergence of open-source data such as Sentinel-1, SAR images' availability and importance in green algae research have gradually increased. For instance, SAR images are essential in calculating green algae's floating and submerged ratio number (FS ratio), which is a good indicator of green tide's growth stage [13]. SAR emits radar pluses and receives backscattering signals from the sea surface, named normalized radar cross section (NRCS). NRCS is usually modulated by the Bragg waves induced by the winds and currents. Green algae float on the sea surface, behave like solid objects, and cause strong volume scattering to the incoming radar signal due to their solid structure. As a result, green

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algae have a prominent texture on SAR images: algae pixels are brighter than seawater, presenting as irregular belts, slicks, or dots [see Fig. 1(b)] [14]. Traditional algae detection method using SAR images is mainly gray-level threshold supported by statistical principles, such as maximum variance or histogram statistics [14], [15]. However, the threshold method is challenging to distinguish dotted green algae from salt-and-pepper noise because of the similar NRCS values. Green algae with small NRCS values are also hard to be extracted accurately.

Deep learning (DL) has been introduced to remote sensing image processing, especially object recognition and detection [16], [17], [18], [19]. A DL-based detection method is an end-to-end approach that automatically integrates different information to achieve feature extraction and data processing. The detection process of DL-based models extracts object features through convolution and pooling layers, breaking the limitation of the threshold method. U-net is a powerful DL model for pixelwise classification and requires fewer training samples [20]. For example, Gao et al. [13] proposed the AlgaeNet based on the U-net framework to detect green algae relying on the NRCS values. However, the texture is also significant information as NRCS provided by SAR images [21] and proved helpful for SAR image classification [22].

Previous studies have revealed that green tide in the Yellow Sea has evident interannual variability [23], [24], which will bring different levels of environmental damage and economic loss. Ulva prolifera is more adaptable than other algal species to higher seawater temperatures in summer [2], [25]. Furthermore, the high uptake rate of nutrients under suitable temperature conditions makes Ulva prolifera germinate and propagate fast [26]. Therefore, seawater temperature and nutrient supply would significantly influence the green tide bloom. Then, what is the relationship between the green tide scales and seawater temperature/nutrients? Laboratory experiments have given the biological nature of green algae for seawater temperature and various nutrients [25], [26], [27], [28]. Due to the characteristics of Ulva prolifera floating on the sea surface, the influence of seawater temperature on Ulva prolifera is generally characterized by sea surface temperature (SST). Therefore, the correlation between the bloom scales and the SST/nutrient distributions has also been analyzed [23], [29], [30]. However, the surrounding seawater temperature and available nutrients should be the practical part affecting the growth of green algae, further forming the interannual variation.

The objectives of this study are three folds: 1) to propose a textural-enhanced DL model for green algae detection; 2) to monitor the interannual variations of the green tide scales in the Yellow Sea from 2019 to 2021; 3) to explore the relationship between green tide interannual variations and available environmental factors.

The remaining article is organized as follows. Section II introduces the study area and the data source. Section III describes the model structure and model performance. Section IV shows the green tide interannual variations in the Yellow Sea from 2019 to 2021 and its relationships with environmental factors. Finally, the conclusions are drawn in Section V.

II. Study Area and Data Used

A. Study Area

The Yellow Sea locates between China and the Korean Peninsula [see Fig. 1(a)]. The western Yellow Sea enjoys abundant illumination in summer, and the SST remains about 25.0 °C. Several rivers and well-developed aquaculture industries in China's eastern nearshore cause higher nutrient concentrations in the western basin [9], [31]. Moreover, aquacultural rafts produce green algae spores in seawater [32]. The SST, illumination, and nutrient environment in summer are suitable for the growth and germination of green algae spores, resulting in the green algae bloom [33]. Proliferated green algae drift northward from the Subei Shoal under the surface current and wind field, finally accumulating along the coast of the Shandong Peninsula [see Fig. 1(a)] [34]. Our study area is shown in the scope of SAR images, which covers the green algae distribution zone in the Yellow Sea.

B. Satellite Images and Textural Features

We collected 119 Sentinel-1 Level-1 ground range detected high-resolution (GRDH) images in the Yellow Sea from 2015 to 2021. Sentinel-1 is a European Space Agency's SAR satellite with VV and VH polarizations and a six-day revisiting period. These SAR images are acquired in interferometric wide (IW) mode with 250-km swath and 10-m resolution [35]. Under such high resolution, Sentinel-1 images can observe tiny algae patches, which are the main pattern in the initial and dissipation stages. Therefore, Sentinel-1 images can observe the relatively complete growth process, and the six-day revisiting period enables Sentinel-1 images to monitor changes in green tide domains in time.

We performed radiometric calibration, speckle filtering, and terrain correction for SAR images using Sentinel Application Platform (SNAP) 8.0 software to get NRCS images in VV/VH channels. The NRCS images in VV/VH channels were converted into decibel versions. Due to the side-view imaging, the NRCS values of SAR image decrease monotonically with the increase of incidence angles, which would interfere with the detection results. Moreover, the empirical linear model is commonly used to correct the incidence angle effects in SAR image classification tasks [36], [37]. Therefore, we extracted the NRCS and incidence angle values along a cross section containing clean seawater pixels and fit a line to compensate for the incidence angle dependency of NRCS values, referencing the image center angle.

Fig. 1(c) shows that green algae are irregular brighter patches on seawater, such as belts, slicks, and dots. These features can be well depicted on the gray-level co-occurrence matrix (GLCM)-based statistical features [38], which have been proved valid to assist the classification of SAR images [22], [39]. These statistical features portray image textural information from various perspectives, and several features are highly correlated [40]. Therefore, we select four commonly used features: mean, angular second moment (ASM), entropy, and correlation to enhance the algae–seawater texture feature contrast [41], [42]. In addition, the proportion of land pixels on the collected SAR image will change the

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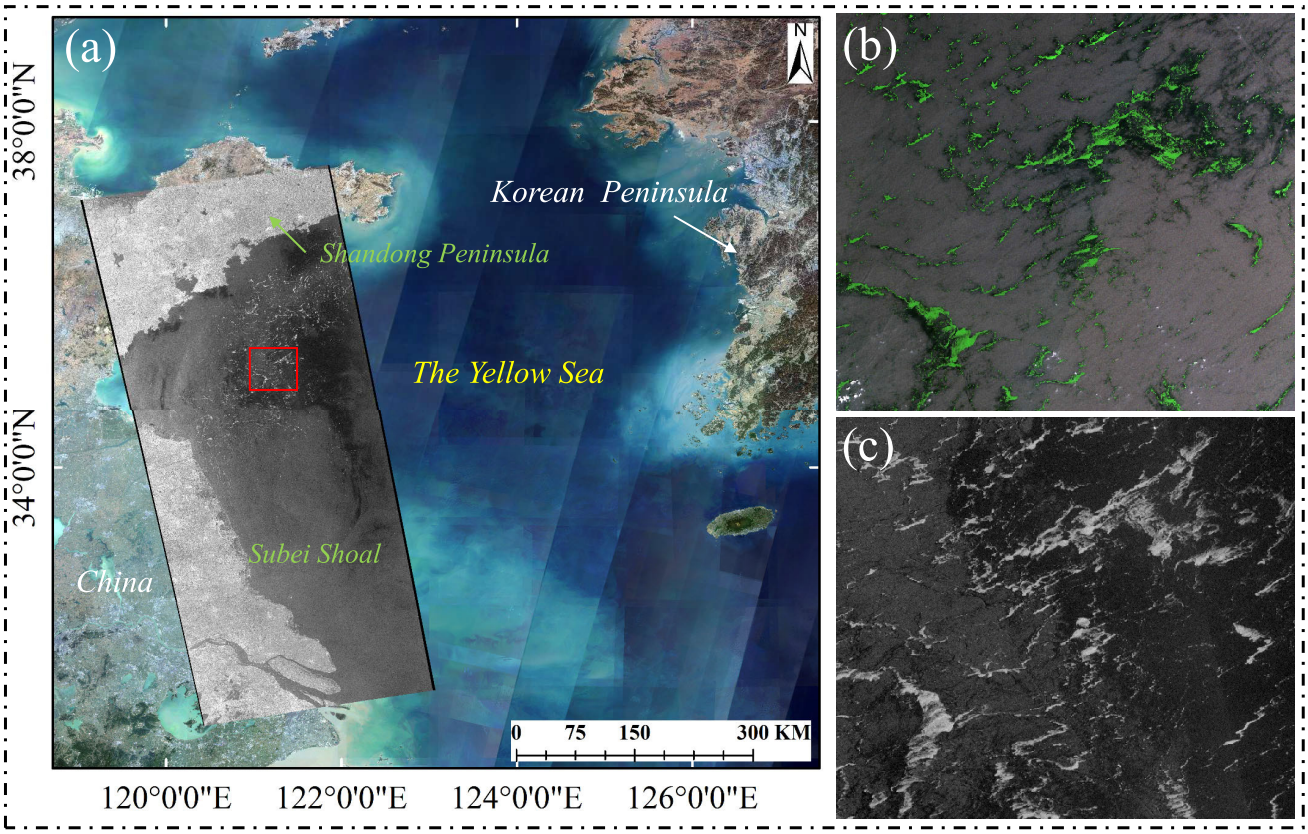


Fig. 1. Study area and green algae signatures on SAR and optical images. (a) Map of the Yellow Sea and our study area. The true-color image is Sentinel-2 (Band 4/3/2), and the rectangle box-bounded greyscale image is the Sentinel-1 VV-polarized image. (b) True-color Sentinel-2 image at the red box of (a) on June 23, 2019. The green patches on the sea surface are green algae. (c) Sentinel-1 VV-polarized SAR image at the red box of (a) on June 23, 2019. Green algae appear as bright patches.

grayscale features of green algae and seawater. Therefore, we masked land before calculating the GLCM features.

III. Methodology

A. Dataset Construction

The wind and wave could undulate the sea surface and bring noticeable NRCS differences of seawater on SAR images. The seawater fluctuation will also change the green algae concentration. Therefore, we selected the images under different sea surfaces and algae-growth conditions to form the dataset, and the other images are for green tide monitoring. Referring to the morphology and position of green algae on Sentinel-2 optical images [see Fig. 1(b)], we made green algae labels on Sentinel-1 VV-polarized images using the annotation tool Labelme software [43]. Pixels are labeled as two types: algae and seawater. All employed images and corresponding labels are clipped into small samples of 128 × 128 pixels. We selected the sample pairs with various typical algae shapes (belt, slick, and dot) to train, verify, and test the proposed DL-based model. Green algae pixels on SAR images are far fewer than seawater pixels; abundant clipped patches without green algae will be processed when we employ the DL model in the green tide monitoring. Therefore, we retained the samples with unremarkable green algae or pure seawater in the testing dataset. Finally, we randomly constructed a training dataset containing 6317 sample pairs (70% for training and 30% for validation) and a testing dataset containing 2124 sample pairs.

B. Model Structure

The classic U-net framework has been widely used in pixelwise classification tasks due to better classification performance and fewer sample requirements [13], [44]. Therefore, we proposed a green algae detection model based on the U-net framework, called GA-Net. The U-Net is named for its almost symmetric encoder–decoder architecture like a "U" shape. The encoder–decoder framework is commonly used in pixelwise classification tasks [13], [16], [18]. Fig. 2(a) shows that the framework combines the input, encoder, decoder, and prediction modules. The encoder module has five stages, except for the first stage with only a convolution block; each of the other encoder stages consists of one max-pooling layer and one convolution block. The decoder path has four stages; each decoder stage comprises one upsampling layer, one concatenation layer, and one convolution block. First, the encoder module extracts downscaled high-level feature maps restored by the decoder module. Next, the intermediate feature maps extracted by the encoder and decoder are connected to form multiscale feature maps for pixelwise classifications. Then, the prediction module with one convolution layer and sigmoid activation function achieves the classification results.

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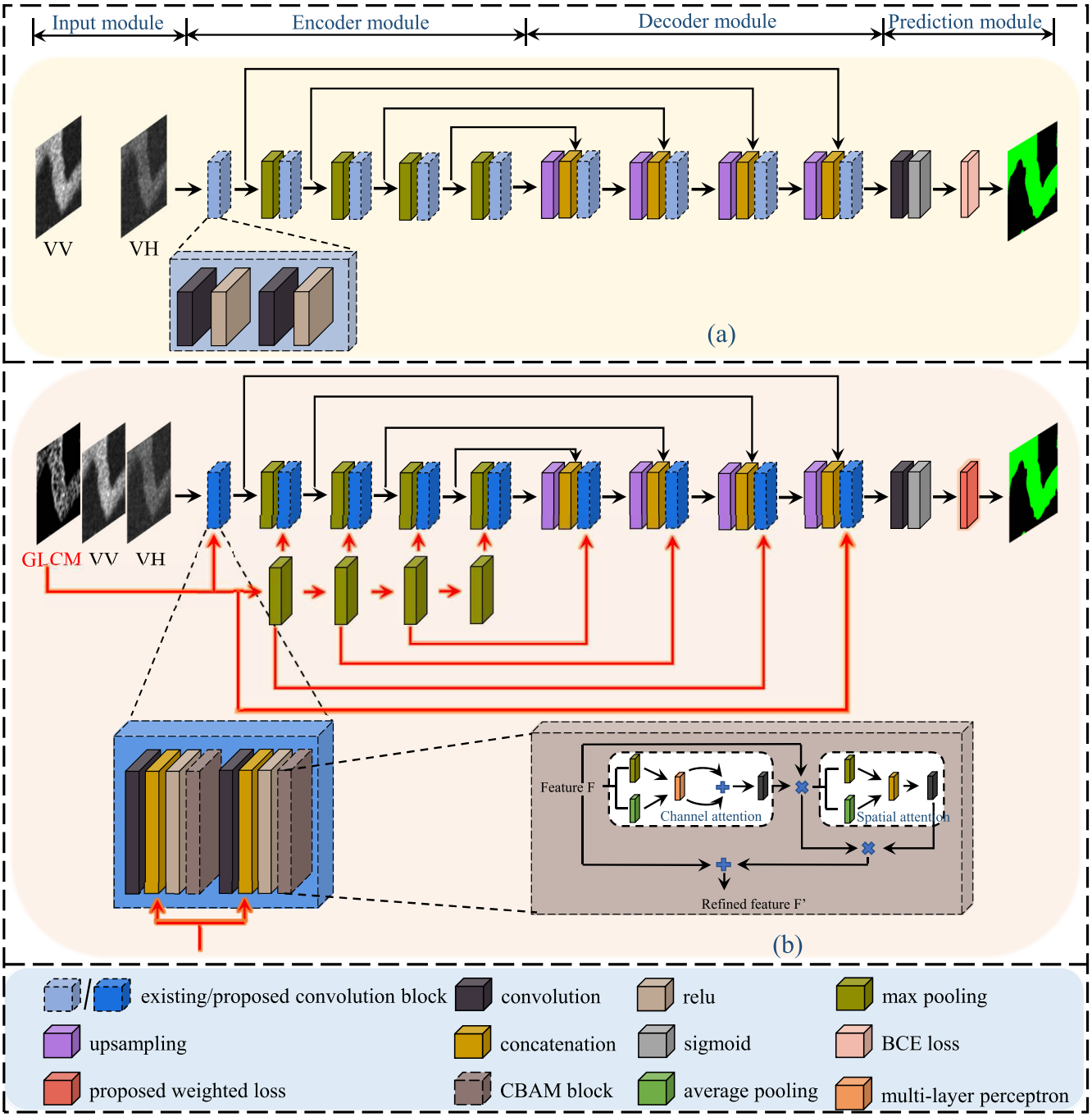


Fig. 2. GA-Net with considering textural features and algae–seawater imbalance based on the U-Net framework. GLCM represents the texture set, including mean, ASM, entropy, and correlation. In addition, there are four tailored modifications: texture-fused input dataset, texture concatenation, CBAM embedding, and weighted loss function. (a) Classic U-net framework. (b) Proposed GA-Net.

To achieve accurate detections for green algae, GA-Net makes four notable improvements based on the original U-Net framework.

First, we adopted a texture-fused input strategy that incorporates the multiphysics-aware information of the SAR image. The input part considers VV-/VH-polarized NRCS and GLCM feature maps (mean, ASM, entropy, and correlation), which provide textural information.

Second, according to the characteristics of green algae, we built a texture-enhanced path called texture concatenation. Green algae with fuzzy boundaries require excellent texture recognition ability. We extracted the GLCM features from the input dataset into pooling layers to generate feature maps with different sizes of 64×64, 32×32, 16×16, and 8×8 pixels. Multiscale GLCM features are concatenated to feature tensors generated by the convolutional layer [see Fig. 2(b)]. The concatenated GLCM texture channels can give a rough reference for the segmentation process due to the more distinct algae–seawater contrast.

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Third, we designed a weighted loss function for SAR images to deal with algae–seawater proportion imbalance. Green algae pixels occupy only up to 14% of the training labels, making DL-based models learn more seawater features than algae ones. The loss function is a criterion to evaluate the bias between the classification results and the corresponding ground truth. The closer the loss function value is to 0, the closer the model prediction result is to the ground truth. Ground truth refers to the manually annotated labels based on the analyst's visual interpretation. The criterion determines the modulation effect of loss function on the training process. Binary cross entropy (BCE) loss measures the probability distribution similarity between the predicted results and ground truth to evaluate the accuracy of prediction results. BCE is commonly used in binary classification tasks because it is more conducive to model convergence. However, BCE is more suitable for the dataset with a balanced proportion. Focal loss (FL) is derived from BCE and suitable in the extremely imbalanced samples by altering two key parameter settings: α for increasing the contribution of the minority category to the loss function and γ for paying more attention to the confused samples (in our model, $\alpha = 0.14$ and $\gamma = 2$) [45]. However, FL is greatly disturbed by label error, which is inevitable in manual labeling. Therefore, we combined BCE and FL with optimal weights ($0.7 \times$ BCE $+ 0.3 \times$ FL) to modulate the detection process.

Fourth, we embedded the convolutional block attention module (CBAM) after each convolution layer [46]. Almost every image sample has fewer green algae pixels than the seawater pixels. Thus, the effective reservation/utilization of features in spatialwise and channelwise during the pooling/upsampling process is significant to the detection result. CBAM comprises one channel attention submodule and one spatial attention submodule. Both use average pooling and max pooling to generate parameters for the input feature tensor F and construct the refined feature tensor F', which would be more discriminative [see Fig. 2(b)].

Besides, some common tricks are integrated into the GA-Net model. Batch normalization [47] and early stopping are used to prevent overfitting and facilitate the adaptation of the DL model to green algae detection. The sigmoid function would obtain the output probability (0, 1) of the green algae pixels: if the pixel value is greater than 0.5, the pixel is taken as green algae; otherwise, it is the background.

C. Model Performance

"Mean intersection over union (mIOU)," "recall," "precision," "accuracy," and "F1-score" are commonly used in pixelwise classification tasks to evaluate the model performance. They are expressed as follows:

$$\text{mIOU} = \frac{1}{k+1}\sum_{i=0}^{k}\frac{\text{T P}}{\text{T P}+\text{F P}+\text{F N}}$$

$$\text{recall} = \frac{\text{T P}}{\text{T P}+\text{F N}}$$

$$\text{precision} = \frac{\text{T P}}{\text{T P}+\text{F P}}$$

$$\text{accuracy} = \frac{\text{T P}+\text{T N}}{\text{T P}+\text{T N}+\text{F P}+\text{F N}}$$

$$\text{F1score} = \frac{2 \times \text{recall} \times \text{precision}}{\text{recall} + \text{precision}}$$

where true positive (TP) and true negative (TN) are the numbers of pixels correctly detected as green algae and seawater, respectively. False positive (FP) and false negative (FN) are the numbers of pixels misclassified as green algae and seawater, respectively. Our detection task aims to detect green algae in the seawater background, and category number k is two.

We evaluated the prediction results based on 2124 pairs of testing samples based on the marked dataset and the proposed GA-Net model. Because green algae detection is essentially a semantic segmentation task in computer vision, previous semantic segmentation models, U-net [20], VGG-16 [48], and AlgaeNet [13], are also trained and tested using the same dataset. The evaluation results of these models are shown in Table I. The GA-Net model achieves the highest scores in "mIOU," "accuracy," "precision," and "F1-score," outperforming the other three segmentation methods. On the other hand, though the VGG-16 model has the highest "recall," the corresponding lowest "precision" indicates that the VGG-16 model has a higher false alarm, overestimating the green tide scale.

Moreover, we examined the model robustness in three typical environment scenarios: the nearshore, low, and high algae aggregation regions. The nearshore region has complicated sea surface conditions close to the shorelines due to high turbidity in shallow water depth and interference from shipping and aquaculture. "Aggregation" refers to the average area of each algae patch in a specific region, i.e., the sum algae area/the number of algae patches. The low algae aggregation region has many tiny algal slicks and the common fuzzy algae–seawater boundaries. Algae in the high algae aggregation region have more distinct algae–seawater boundaries, which are easier to be classified on SAR images. Table I and Fig. 3 give the detection results of four models in the above three typical environmental regions (1280×1280 pixels) randomly selected on June 30, 2021. U-net has the highest "mIOU," "accuracy," "precision," and "F1-score" in the nearshore region. Compared with VGG-16 and U-Net, AlgaeNet and GA-Net can better distinguish the green algae from ships, and GA-Net has more vital anti-interference ability, as shown in red boxes of Fig. 3(b)–(f). Green algae are brighter in the low algae aggregation region than in the nearshore region. The GA-Net model achieves the "mIOU" of 81.91%, outperforming the other three models, especially in the fuzzy algae–seawater boundaries of Fig. 3(g)–(k), red boxes. In the high algae aggregation region, all four detection models perform well. The GA-Net model gains the highest "mIOU," "accuracy," "precision," and "F1-score". Each detection model has no apparent differences in the detection results from Fig. 3(l)–(p). The better detection performance of GA-Net is due to the more accurate identification of the fuzzy algae–seawater boundaries. Therefore, the comparison in three regions demonstrates that the GA-Net model can distinguish green algae and ships well and precisely identify the algae–seawater boundaries.

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TABLE I
EVALUATION INDICATORS OF DIFFERENT DETECTION METHODS

	Detection model	mIOU	Accuracy	Precision	Recall	F1-score
Randomly selected patches	VGG-16	81.53%	97.65%	87.36%	**92.43%**	89.82%
	U-net	83.45%	97.98%	90.94%	91.02%	90.98%
	AlgaeNet	83.54%	98.02%	92.76%	89.37%	91.03%
	GA-Net	**86.31%**	**98.36%**	**93.29%**	92.03%	**92.65%**
Nearshore region	VGG-16	67.11%	93.30%	70.03%	94.14%	80.32%
	U-net	67.24%	93.14%	68.68%	**96.99%**	80.41%
	AlgaeNet	71.75%	94.60%	74.87%	94.51%	83.55%
	GA-Net	**74.93%**	**95.37%**	**77.84%**	95.26%	**85.67%**
Low algae aggregation region	VGG-16	71.64%	93.55%	74.22%	95.38%	83.48%
	U-net	79.78%	95.93%	84.02%	94.05%	88.75%
	AlgaeNet	81.40%	96.33%	**85.84%**	94.02%	89.75%
	GA-Net	**81.91%**	**96.39%**	85.07%	**95.67%**	**90.06%**
High algae aggregation region	VGG-16	91.63%	99.15%	92.96%	98.46%	95.63%
	U-net	89.77%	98.92%	90.15%	**99.54%**	94.61%
	AlgaeNet	93.67%	99.37%	94.98%	98.55%	96.73%
	GA-Net	**94.47%**	**99.45%**	**95.54%**	98.83%	**97.16%**

To further verify the performance of the GA-Net model in the entire image with extensive algae coverage in the above three environmental scenes during the green tide bloom, we randomly selected a SAR image on June 23, 2019. Fig. 4(b) shows that although a few artificial objects along coastlines are misidentified as green algae, the detection result in the open seawater is accurate. The detected algae pattern is consistent with the distribution pattern of the original image. Furthermore, we enlarged one typical gulf region [the red box of Fig. 4(a) and (b)] where the "detected algae" are obvious. This gulf region has no green algae in this image. The "detected algae" along the coastlines are artificial facilities (e.g., aquaculture zone) with similar NRCS features to green algae on SAR images. We also performed three other trained models in this region. The green algae misclassification ratio, defined as the number of pixels misclassified divided by the number of total unmasked pixels, is calculated to quantify the performance of each model. Experiment results show that GA-Net has the lowest misclassification ratio of 14.25%, indicating the model's superiority.

D. Ablation Experiments

The GA-Net model is based on the classical U-net framework with four improvements. This model has high accuracy for green algae detection, especially in fuzzy edges. However, is every modification effective? Which improvement contributes more? Therefore, inspired by [49], we designed four experiments to demonstrate the contributions of each improvement.

First, we set the GA-net model as the control group and moved GLCM texture features in the input dataset as Exp. 1 to examine the function of texture maps fused in the input dataset. Then, we moved the concatenation design, CBAM block from the GA-Net model as Exp. 2 and Exp. 3. Finally, we substituted the BCE loss for the hybrid weighted loss in Exp. 4 to learn the effect of our new loss function. All the ablation experiments have the same hyperparameters and overfitting suppression strategies as the control group. The lower the evaluation indicators in the ablation experiments, the more critical the modification. Table II shows that our proposed GA-Net model has the highest "mIOU," "accuracy," "recall," and "F1-score," indicating that each modification effectively improves the green algae detection performance. Moreover, from the perspective of "mIOU," "recall," and "F1-score," the ranker of the evaluation indicator scores is Exp. 4 < Exp. 2 < Exp. 1 < Exp.3, implying that the contribution of each modification is: the weighted loss function > concatenation design > texture fused input dataset > CBAM embedding.

IV. RESULTS

A. Green Tide Scales From 2019 to 2021

The proposed GA-Net model provides technical support for accurately detecting green tide bloom. Therefore, we employed the GA-Net model to detect the process of the green tide during 2019–2021 based on the collected SAR images. The interannual variations of green tide are characterized by bloom duration (days), maximum coverage area, and nearshore damage to the southern coastlines of the Shandong Peninsula.

Bloom duration is one of the critical indicators for estimating the green tide scales. Fig. 5(a) shows the green algae bloom from 2019 to 2021. In 2019, the bloom of green algae lasted for 96 days, and the algal bloom peak (maximum coverage) appeared on June 23. The mean growth

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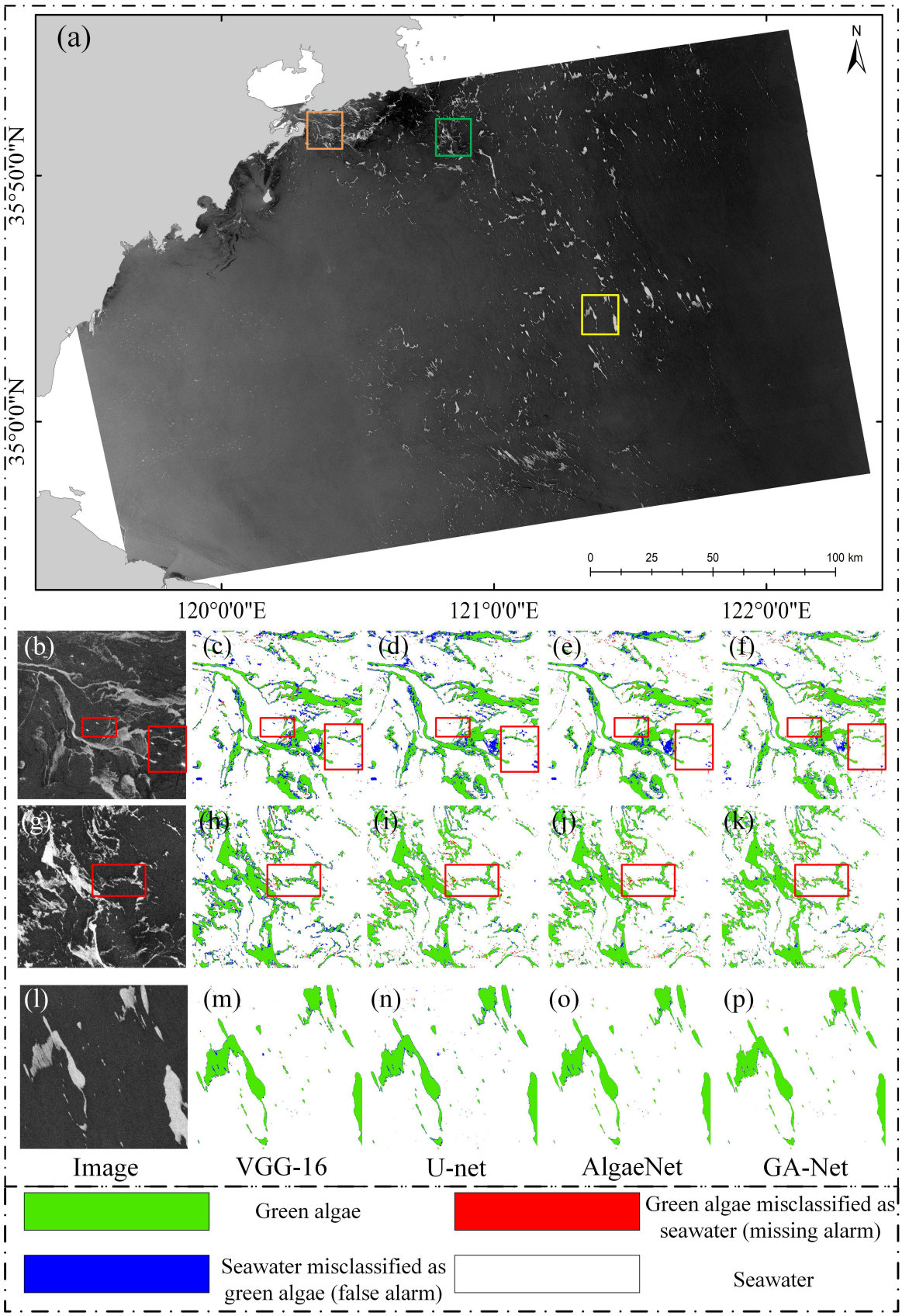


Fig. 3. Randomly selected testing images and detection results of the GA-Net and previous advanced detection models in three different conditions. (a) VV-polarized image with Universal Transverse Mercator (UTM) projection on June 30, 2021. The orange box represents a nearshore region. (b)–(f) Larger vision and corresponding detection results. The green box shows a low-aggregated green algae region. (g)–(k) Details of the image and detection results. Yellow box and (l)–(p) exhibit the images and corresponding detection results of a high-aggregated green algae region.

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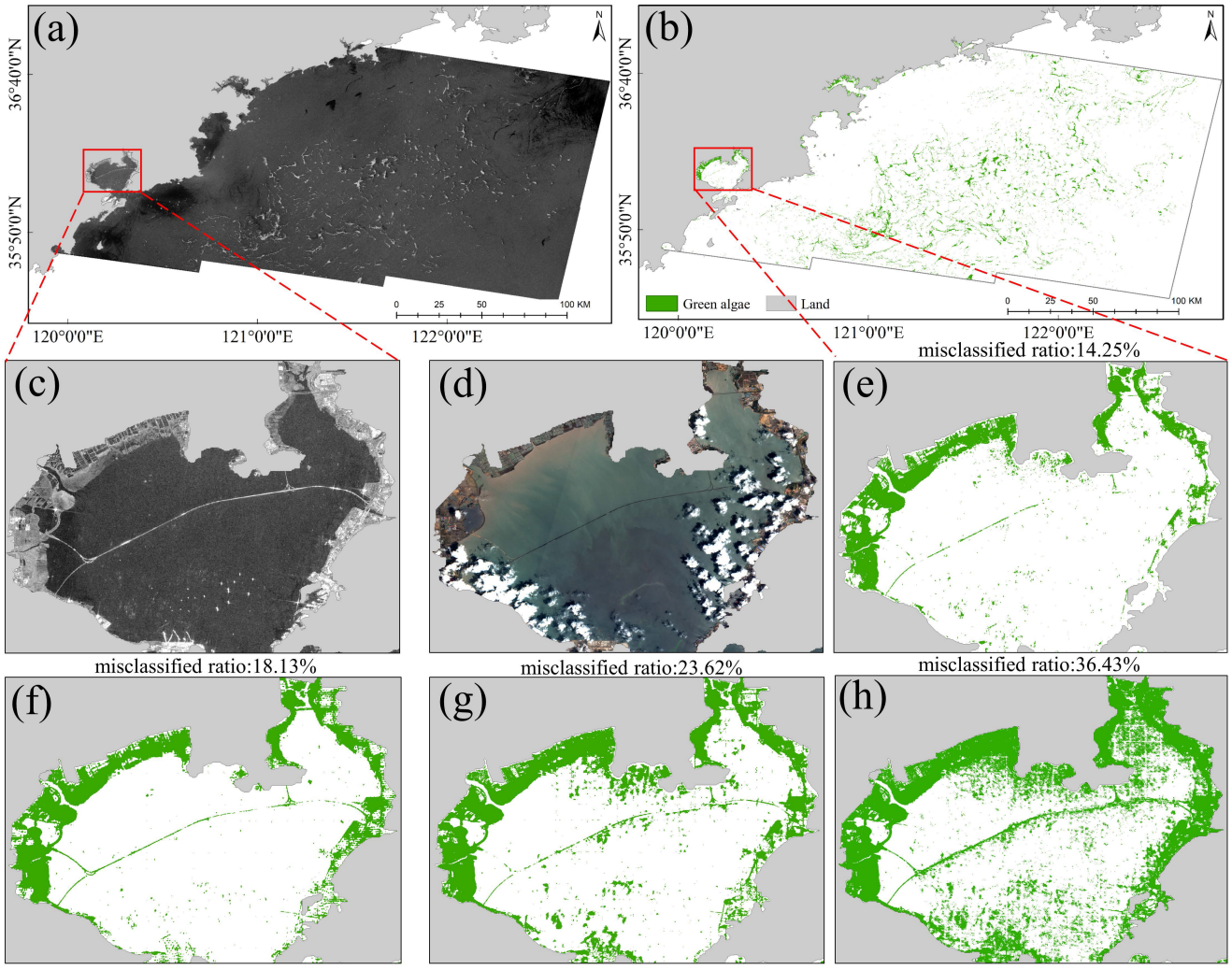


Fig. 4. Model performance on one SAR image on June 23, 2019. (a) VV-polarized SAR image with UTM projection. (b) Detection result of the GA-Net model. (c) and (d) Larger vision of one typical gulf region on SAR/optical image. The optical Sentinel-2 image was acquired on June 26, 2019. (e)–(h) Detection result of this gulf region based on GA-Net/VGG-16/AlgaeNet/U-net.

TABLE II
EVALUATION INDICATORS OF ABLATION EXPERIMENTS

Experiment	mIOU	Accuracy	Precision	Recall	F1-score
Exp. 1	85.74%	98.32%	**94.60%**	90.15%	92.32%
Exp. 2	85.35%	98.27%	94.24%	90.04%	92.10%
Exp. 3	85.84%	98.31%	93.51%	91.28%	92.38%
Exp. 4	83.67%	98.07%	94.49%	87.96%	91.11%
Control group (GA-Net)	**86.31%**	**98.36%**	93.29%	**92.03%**	**92.65%**

rate during the area-increasing period was 74.8 km^2/day. In 2020, green algae reached a bloom peak on June 11, lasting only 42 days. The area-increasing period in 2020 had a flat growth rate of 19.3 km^2/day. Compared with the previous two years, algae appeared earlier in 2021. The bloom duration was 84 days. June 17 was the peak-bloom day, and the mean growth rate was 47.7 km^2/day before the bloom peak. In brief, 2019 (2020) has the longest (shortest) bloom duration and biggest (smallest) growth rate in the area-increasing period.

The maximum coverage area is commonly used to estimate the green tide scales [50]. Fig. 5(c)–(e) shows the algae distributions in the maximum coverage days during 2019–2021. In 2019, the maximum coverage area was 2293.37 km^2, filling the entire study region: green algae accumulated in open ocean areas and the Subei Shoal (see Fig. 1); the northern nearshore

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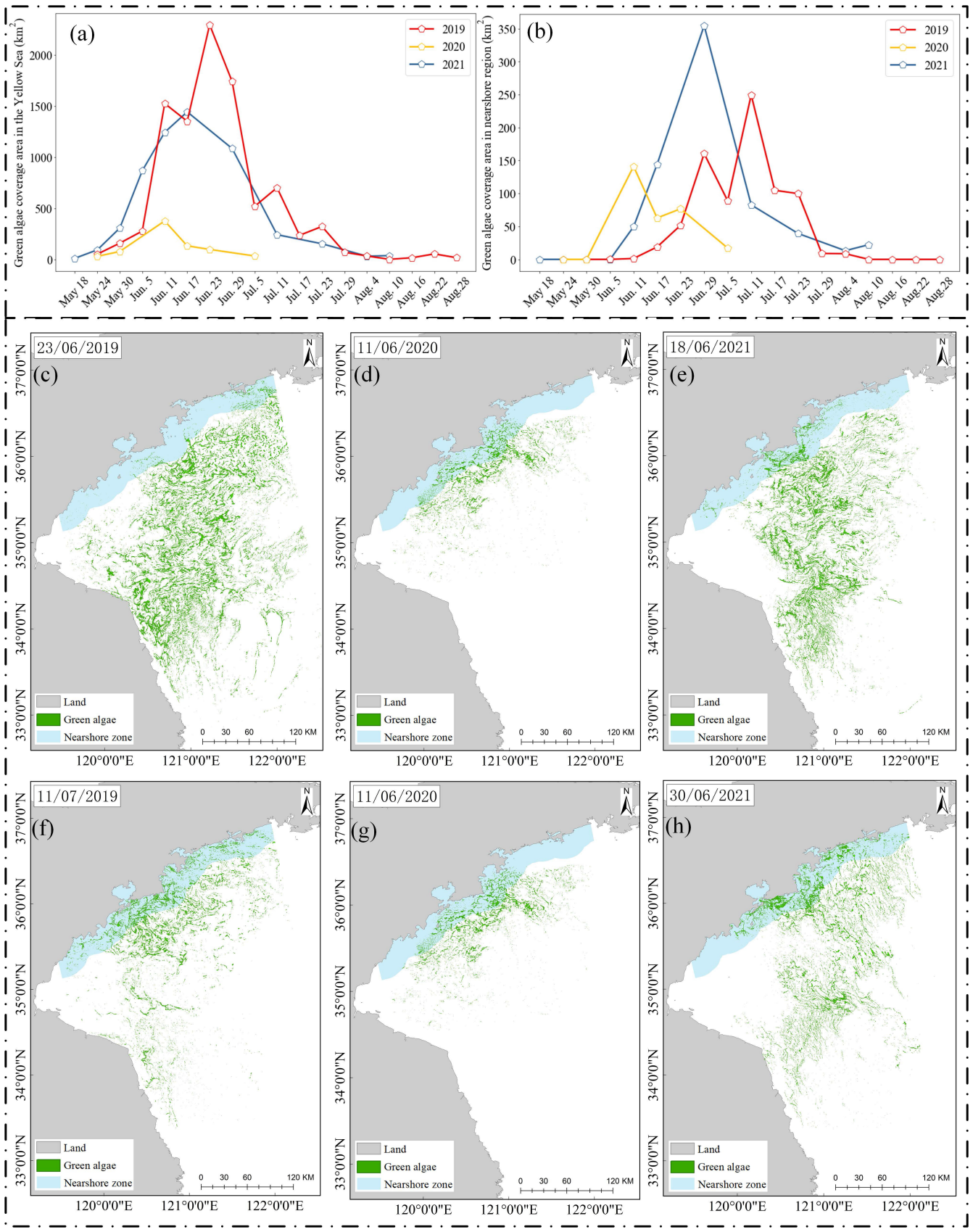


Fig. 5. Coverage area in (a) Yellow Sea and (b) nearshore zone. (c)–(e) and (f)–(h) Algae (green patches) distribution on the maximum coverage area day in the whole research area and the nearshore zone, respectively. For the convenience of drawing, the images acquired time of (a) and (b) in 2021 is one day in advance. Blue shadow is the scope of 12 nautical miles off the coastlines, i.e., the nearshore zone.

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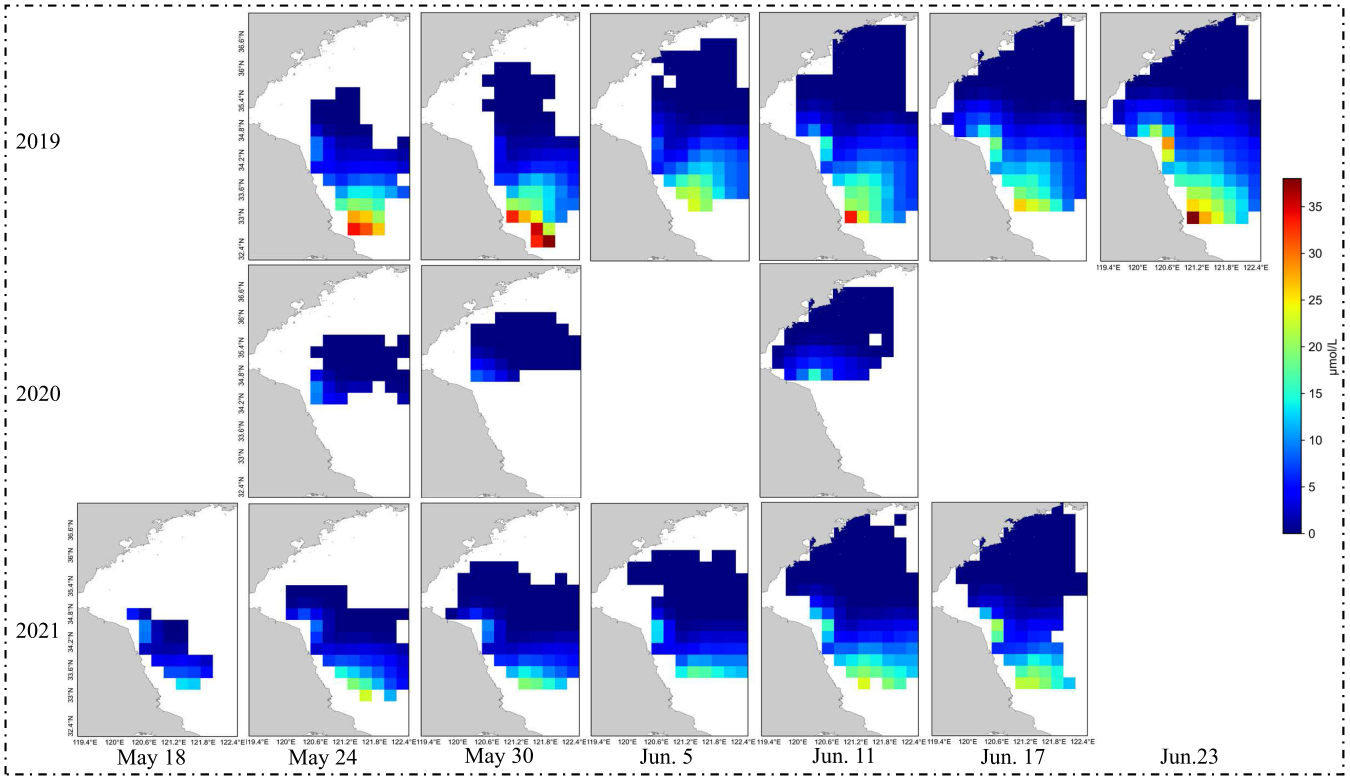


Fig. 6. Nitrate concentration in the green algae distributed regions during the area-increasing period. For the convenience of drawing, the images acquired time in 2021 are one day in advance.

zone was barely affected. The maximum coverage area in 2020 was 375.21 km^2, mainly distributed near the southern coastlines of the Shandong Peninsula. The maximum coverage area in 2021 was 1441.96 km^2. Green algae were distributed as belts, connecting the Subei Shoal and the northern coastal. Therefore, zonal distribution was more compact compared with 2019. In short, the maximum coverage area in 2019 is the biggest, about six (one and a half) times 2020 (2021).

Nearshore damage, especially the damage to the southern coastlines of the Shandong Peninsula, is a concern in the algae bloom period of the Yellow Sea. Twelve nautical miles off the coastline [shown as the blue areas in Fig. 5(c)–(h)] covers the main human-activity domains in the sea, such as aquaculture and tourism. Therefore, we counted the green algae coverage area in this nearshore zone to measure the nearshore damages. Fig. 5(b)–(h) shows that green algae attacked the southern coastlines of the Shandong Peninsula most severely on July 11, 2019, occupying 249.54 km^2 in the nearshore zone. In comparison, the invasion area of the green tide was 141.10 (354.78) km^2 on June 2020 (2021). In conclusion, the biggest (smallest) nearshore damage to the southern coastlines of the Shandong Peninsula is in 2021 (2020).

B. Relationship Between the Interannual Variation of Green Tide Scales and Nutrient/SST

Seawater eutrophication is mainly due to human activities such as domestic sewage and fertilizers from agricultural fields and often occurs in nearshore waters. Land surface runoff carries nutrients produced on land to nearshore, and rainfall raises the nutrient concentration of nearshore seawater by scouring the land and increasing the runoff flow [51]. Therefore, the intensity of rainfall and the flow of runoff affect the nutrient concentration of nearshore seawater, which interferes with algae's reproduction rate and outbreak scale [52]. Li et al. [53] confirmed that total precipitation was positively correlated with the maximum daily green algae coverage, and Xiao et al. [24] found a positive correlation between a river flow and the green tide's maximum biomass. However, both total precipitation and runoff flow can only partially represent the nutrient concentration of seawater, and various nutrient elements have different effects on the growth and reproduction of green algae [30], [31], [54].

Higher concentrations of nitrate and phosphate in eutrophic water provide the foundation for the green tide [55]. Green algae uptake more nitrogen, and the increase in phosphorus concentration makes a very slight promoting effect. The germination and growth of green algae mainly rely on nitrogen concentration [54], [56]. Most fixed inorganic nitrogen in the ocean is stored in nitrate, and nitrate is the significant material for green algae growth [30]. Therefore, we acquired the daily nitrate concentration data (GLOBAL_ANALYSIS_FORECAST_BIO_001_028) with $0.25° \times 0.25°$ resolution from the Global Monitoring and Forecasting Centre (CMEMS) to explore the relationship between interannual bloom variation and nitrate concentration.

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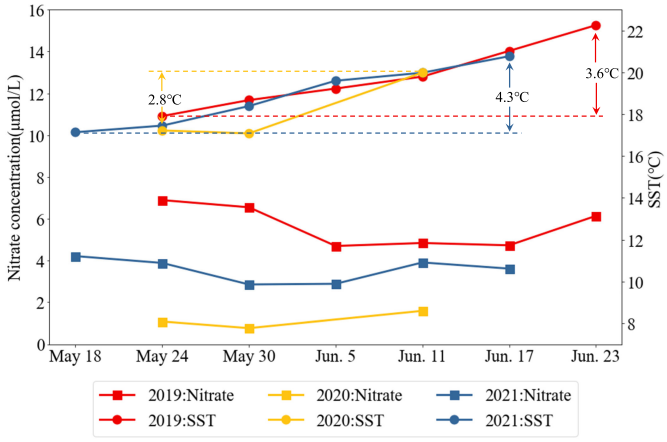


Fig. 7. Mean nitrate concentration and mean SST in the green algae distributed regions during the algal-growth period. For the convenience of drawing, the images acquired time in 2021 are one day in advance.

This product has ten vertical layers and daily intervals produced through Pelagic Interactions Scheme for Carbon and Ecosystem Studies volume 2 (PISCES-v2). Nitrate concentration in the product has good agreement with both World Ocean Atlas Climatology and BGC-Argo data on the global scale [57]. Therefore, based on this dataset, we could estimate the available nitrate supplement for green algae.

As shown in Fig. 5(a), the green algae bloom combines a rapid area-increasing period and an area-decreasing period. Green algae growth in the area-increasing period mainly dominates green tide bloom scales. Moreover, the nutrient concentration near Subei Shoal is higher than in other Yellow Sea regions due to fertilizer for aquaculture and overland runoff [9], [31]. The distribution region of green algae directly affects nutrient utilization. Therefore, we plotted the nitrate distribution and measured mean nitrate concentration in green algae-distributed regions on each imaging day during the area-increasing period to assess the available nitrate for green algae growth (see Figs. 6 and 7). Fig. 6 shows that the broadest distribution range occurs in 2019 during the whole area-increasing period. Green algae had more available nitrate than the other two years in the south of 35°N, resulting in a higher mean nitrate concentration in algae-distributed regions, as shown in Fig. 7.

On the contrary, the distribution region in 2020 is the smallest, and the location is further north, resulting in less available nitrate in the growth period. Fig. 7 also shows that the mean nitrate concentration in algae-distributed regions in 2020 is lower than that in 2019 and 2021; the highest mean nitrate concentration in algae-distributed regions is in 2019. Moreover, the tendency of available nitrate concentration in the algae-distributed regions is the same as that of the maximum green algae coverage areas in 2019–2021, as shown in Figs. 5(a) and 7. Previous studies have revealed that green algae growth positively correlated with dissolved inorganic nitrogen concentration [58]. Therefore, the interannual variation of the green tide in these three years may be partly determined by the available nitrate concentration.

SST is another crucial factor for the green algae bloom. Therefore, we also acquired the SST data from the Operational SST and Sea Ice Analysis (OSTIA) product (SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001) on CMEMS. This dataset is derived from in situ observations and satellite resources with a spatial resolution of 0.05° × 0.05° and a daily interval [59]. The OSTIA SST has been proven to have the smallest root-mean-square difference (RMSD) compared to independent buoy and ship observations in China's coastal and shelf seas [60]. Like nutrient data processing, we measured the available SST within the algae distributed ranges and found the mean SST ranges from 17.0 °C to 23.0 °C within the suitable temperature range (14.0 °C–27.0 °C) for green algae growth (Fig. 7) [25].

Moreover, we also calculated the variation of available SST by subtracting the mean available SST of the first appearance day from the bloom day. We found that the variation of available SST in the algae-distributed regions is 4.3 °C/2.8 °C/3.6 °C in 2019/2020/2021, respectively, which has the same trend as the maximum coverage area in the corresponding year (see Figs. 5(a) and 7). Li et al. [53] also found that the green algae coverage area is positively correlated with the variation of SST, which indicates that the interannual variation in green tide bloom scales may also be partly correlated with the variation of available SST.

Besides the above two predominate environmental factors, the interannual variation of green tide scales is also affected by human interference: updating aquaculture rafts material, controlling the time to recycle rafts, spraying inhibitors, salvaging time and position, and so on. [50], [61], [62]. Unfortunately, these human activities' effects are hard to be quantified.

V. Discussion

Model interpretability and the fusion of DL and prior knowledge (or physical mechanism) are hot issues in the intersection research of geosciences and DL at present [63], [64], [65], [66]. Our research has some contributions to the above two aspects:

A. Fusion of DL and Prior Knowledge (or Physical Mechanism)

The object extraction from SAR images based on DL models usually takes the NRCS images as the gray photographs. The data-driven model relies on the images and labels to complete the extraction task [13], [16], [18], [67]. However, DL models combined with prior knowledge (or physical mechanisms) can reduce the dependence on labeled samples and achieve better results [64]. Based on the classic semantic segmentation U-Net model, this article injected prior knowledge to induce the learning process of the model, which is reflected in the following four improvements: 1) fuse texture features with NRCS information as input; 2) inject the texture features into the convolution layer of each stage to improve the model detection ability at the fuzzy algae–seawater boundaries; 3) embed CBAM module to select discriminative feature channels and regions in each processing stage; 4) set the parameters of the loss function according to the proportion of

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algae and seawater pixels in the training samples. The above four modifications inject two key prior knowledge, i.e., texture features and algae/seawater sample proportion, to the DL model, modulating the training process.

B. Model Interpretability

The DL model interpretability can be divided into self-explanatory and post-explanatory [66]. Self-explanatory refers to embedding physical theoretical rules into the data-driven model to increase the model interpretability. Post-interpretation means that after the model is trained, the role of the submodule is explained by simple machine learning methods (e.g., decision tree) or visualization methods (e.g., grad-CAM) [68], [69]. This study used ablation experiments to analyze the contribution of each model improvement to green algae detection, which is a post-model interpretation method.

VI. Conclusion

Green tide often damages China's Yellow Sea coasts and marine ecology. Remote sensing images, particularly SAR images, which can penetrate the cloud, have become increasingly important in green algae detection. Previous DL-based green algae detection models only consider the NRCS information but not texture information. To this end, this study proposed a texture-enhanced GA-Net model to detect green algae using Sentinel-1 SAR images. We designed four modifications based on the DL segmentation model U-net.

1) Combine texture-fused input datasets to provide multiple information for green algae detection.
2) Concatenate GLCM features after each convolution layer to enhance the texture of algae–seawater.
3) Construct a hybrid loss function to mediate the imbalanced training samples.
4) Embed the CBAM to help the detection process find effective features.

The GA-Net model is trained on the Sentinel-1 SAR images in the Yellow Sea and achieves the "mIOU" of 86.31%, better than classic DL models: VGG-16 of 81.53%, U-net of 83.45%, and AlgaeNet of 83.54%. According to the ablation experiments, the above modifications effectively improve the detection performance, and the weighted loss function contributes more than the others.

We also used the GA-Net model to investigate green algae distribution characteristics in the Yellow Sea from 2019 to 2021. Investigation results show that the bloom durations in 2019, 2020, and 2021 are 96, 42, and 84 days, respectively; the maximum coverage area in the study area of 2019 is 2293.37 km^2, about six (one and a half) times as large as 2020 (2021); and the maximum coverage area in the nearshore region of 2021 is 354.78 km^2, about one and a half (two and a half) times as large as 2019 (2020). Therefore, green algae in 2019 have the most extended bloom duration and the most extensive coverage area. However, the most severe nearshore damage was in 2021. Our analysis found that such interannual variation of green tide scales is partly related to the available mean nitrate concentration and the variation of available SST in algae-distributed regions. For example, less nitrate and smaller SST variation in the algae-distributed regions contributed to the shortest bloom duration and smallest coverage scale. Moreover, human activities, such as spraying inhibitors, updating aquaculture procedures, and salvaging, would also contribute significantly to the small bloom scale in 2020.

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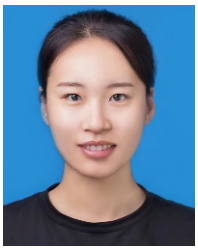
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